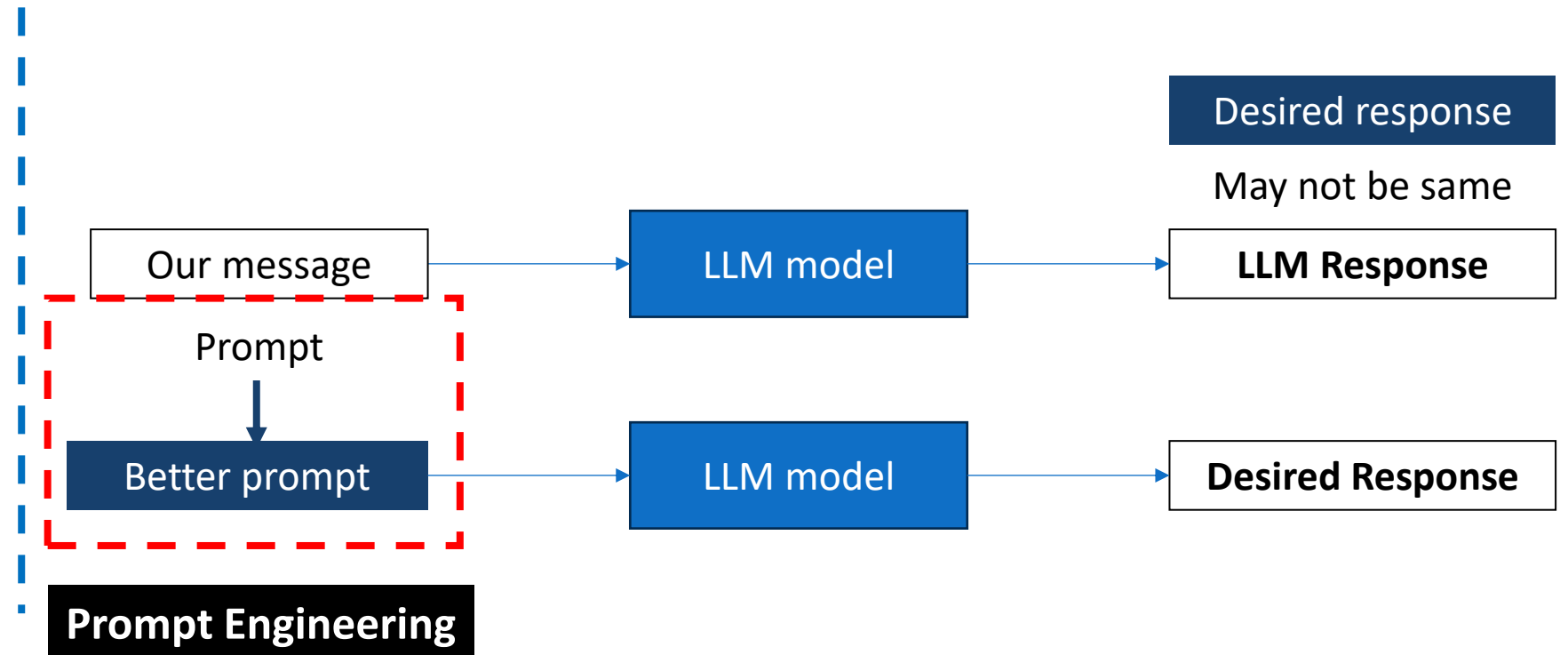


Prompt Engineering

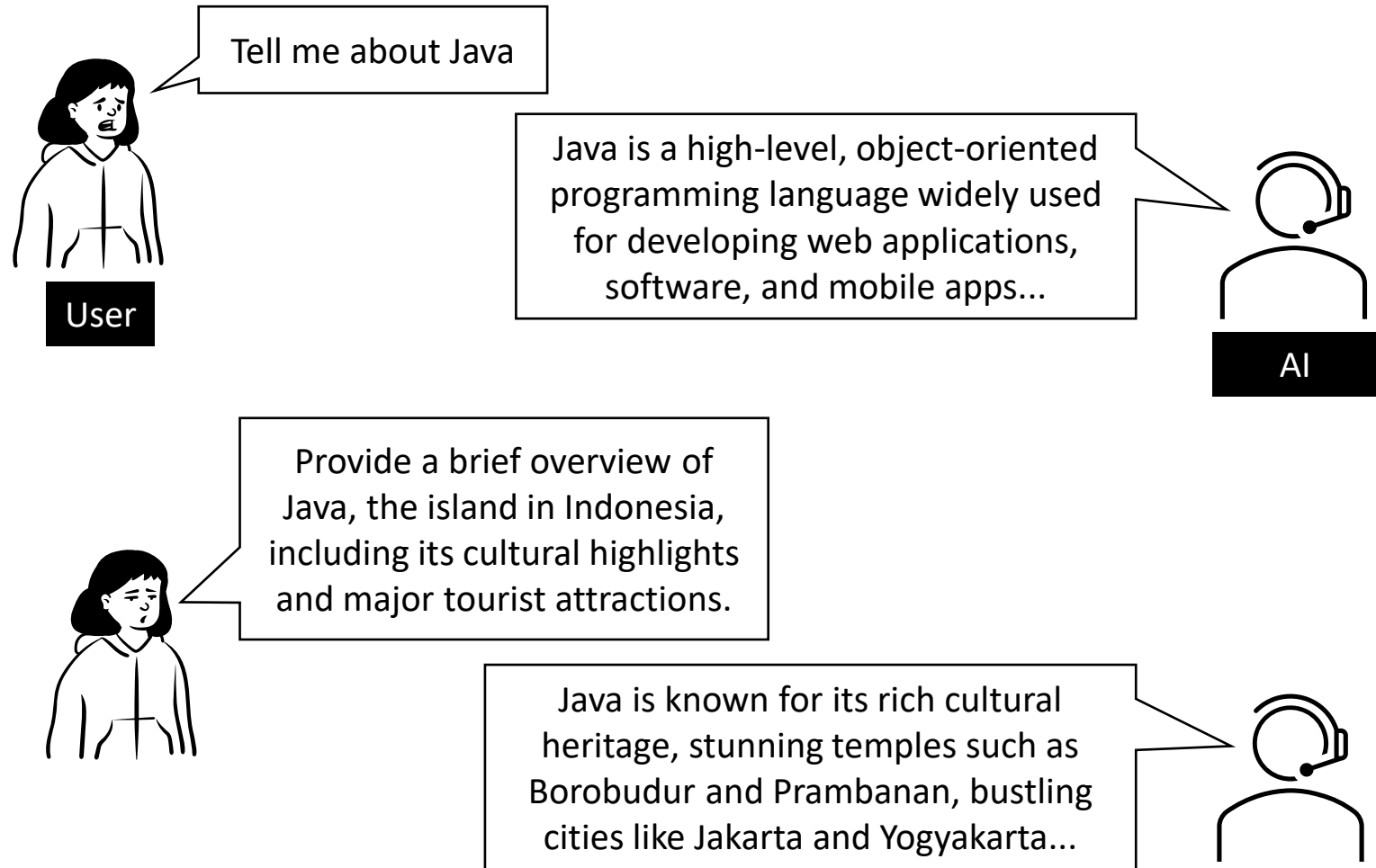
When we want to generate text, we have some desired output text in mind

What



Prompt Engineering

Example



Prompt Engineering

Order of learning

Prompt engineering
basics and best practices

Clearly communicating “WHAT”
we want in the response

Prompting techniques and
frameworks

Communicating “How” to get
the response that we want

Prompt Hyperparameters
and their tuning

Adjusting some controls to
improve response quality

Prompt Evaluation

Assessing the prompt quality



Key to Good Prompting

How to make a prompt detailed and specific both?

Detailed and Specific

Length of response

Tabular,
downloadable etc.

Output format

Target Audience

Style & tone

funny, serious, sad,
angry tone etc.

Examples of output

Specific

- One Paragraph
- 3 lines
- ...

Key to Good Prompting

Detailed and Specific

Detail Type	Why It's Important
Task Objective	Clarifies what output is expected (e.g., summary, code, explanation).
Context/Background	Provides necessary information for accurate and relevant responses.
Format Requirements	Ensures the output matches the desired structure (e.g., table, list, JSON).
Tone or Style	Helps generate output in the right voice (e.g., formal, friendly, technical).
Audience Type	Tailors content complexity and language for the intended reader or user.
Length Constraint	Controls verbosity (e.g., 1 paragraph, 100 words, bullet points).
Examples (if any)	Demonstrates the desired pattern, improving accuracy and consistency.
Constraints or Do's/Don'ts	Prevents unwanted output or ensures inclusion of specific details.



Prompting Best Practices

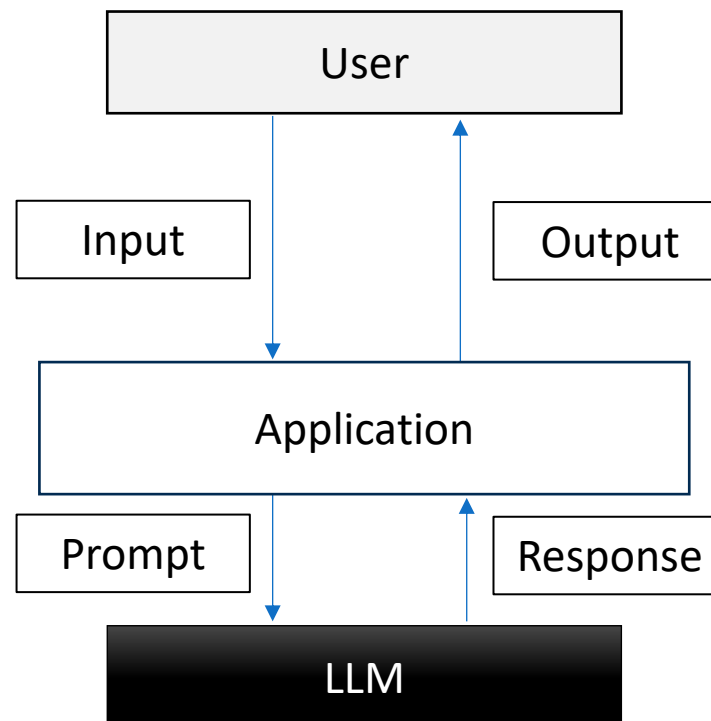
Proven principles

1. Separate instructions and context using delimiters
2. Use Positive prompts
3. Provide steps for the desired outcome
4. Break large tasks into smaller sections
5. Give examples – Few shot prompting



Prompt Template

What?



Prompt

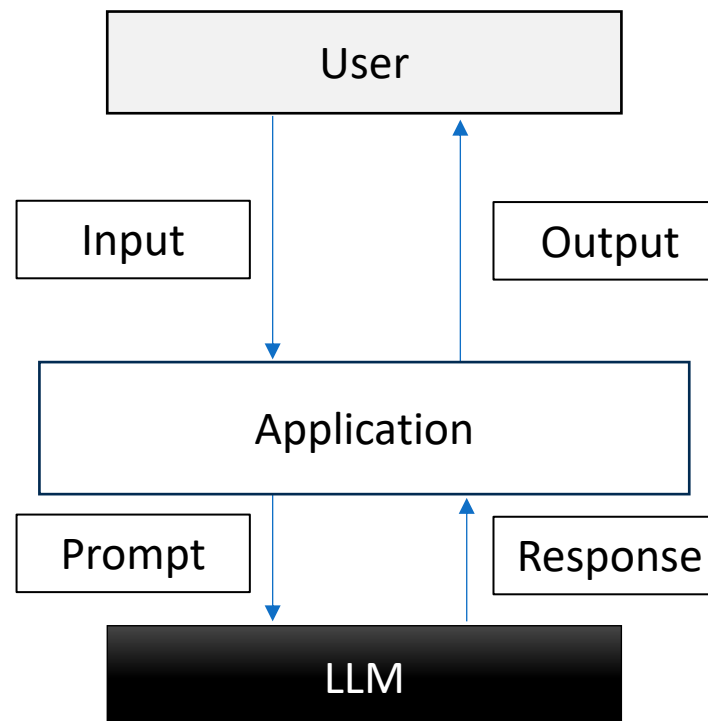
Design a logo for a modern tech startup. The logo should feature a minimalist design with clean lines and a geometric shape. Use a color scheme of blue and white.

Prompt Template

Design a logo for a [type of company]. The logo should feature a minimalist design with clean lines and a geometric shape. Use a color scheme of [Color scheme].

Prompt Template

Why?



What

Prompt templates convert user input to a useful Prompt

Why

Saves time and effort
Enrich input by adding instructions, examples and questions

Prompt Template

Structure

```
from openai import OpenAI
client = OpenAI()

completion = client.chat.completions.create(
    model="gpt-4.1",
    messages=[
        {"role": "assistant", "content": "You are a helpful assistant."},
        {"role": "user", "content": "Hello!"}
    ]
)

print(completion.choices[0].message)
```



Prompt Template

Variables

```
from langchain_core.prompts import ChatPromptTemplate

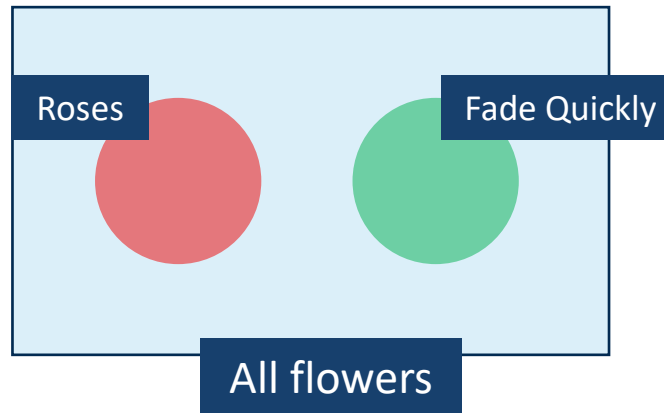
template = ChatPromptTemplate([
    ("system", "You are a helpful AI bot. Your name is {name}."),
    ("human", "Hello, how are you doing?"),
    ("ai", "I'm doing well, thanks!"),
    ("human", "{user_input}"),
])

prompt_value = template.invoke(
    {
        "name": "Bob",
        "user_input": "What is your name?"
    }
)
```

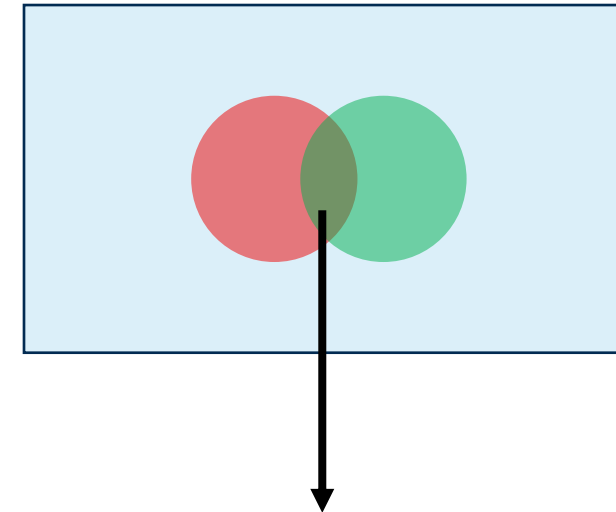
Prompting techniques

Q. If all roses are flowers and some flowers fade quickly; can we say that some roses fade quickly? **Ans. NO**

Chain-of-Thought prompting



This represents that roses don't fade quickly



This represents that some roses fade quickly

Chain-of-Thought prompting

Why

Straight forward problems

Q. What is the capital of India?

A. New Delhi

Q. What are the 7 P's of marketing and how do I apply them on my product?

A. The 7 P's, also known as the extended marketing mix, are a set of strategic tools used by businesses to create a comprehensive marketing plan...

Reasoning problems cannot be answered directly, they need a logical step-by-step approach

Chain-of-Thought (CoT) prompting is used to instruct the LLM to use a logical step-by-step approach while generating the response



Chain-of-Thought prompting

Prompts

Prompt 1: If all roses are flowers and some flowers fade quickly; can we say that some roses fade quickly?

Prompt 2: I purchased a \$20 decorative item. I spent \$6 to repaint it. Then I sold it to the customer for \$32.99. I paid a tax on profit at a rate of 25%. First give me my net gains then show the calculations. Calculation should be after the answer.

Chain-of-Thought prompting

Zero-shot CoT

Simply add “**Let’s think step-by-step**” at the end of the prompt

Prompt 1: If all roses are flowers and some flowers fade quickly; can we say that some roses fade quickly? Let’s think step-by-step.

Prompt 2: I purchased a \$20 decorative item. I spent \$6 to repaint it. Then I sold it to the customer for \$32.99. I paid a tax on profit at a rate of 25%. First give me my net gains then show the calculations. Calculation should be after the answer. Let’s think step-by-step.



Chain-of-Thought prompting

Few-shot CoT

Give examples of step-by-step approach and then ask the question

Prompt:

Q: Jake ran 3 miles on Monday, 5 miles on Wednesday, and 7 miles on Friday. If his weekly goal was 20 miles, how many more miles does he need to run?

A: Jake ran $3 + 5 + 7 = 15$ miles. His goal is 20 miles. $20 - 15 = 5$. The answer is 5.

Q: A factory produced 250 toys in the first week and 370 toys in the second week. If 180 toys were sold, how many toys remain?

A: The factory made $250 + 370 = 620$ toys in total. It sold 180, so $620 - 180 = 440$. The answer is 440.

Q: Sarah had 120 candies. She gave 25 to her brother and then bought 40 more. How many candies does she have now?

A:

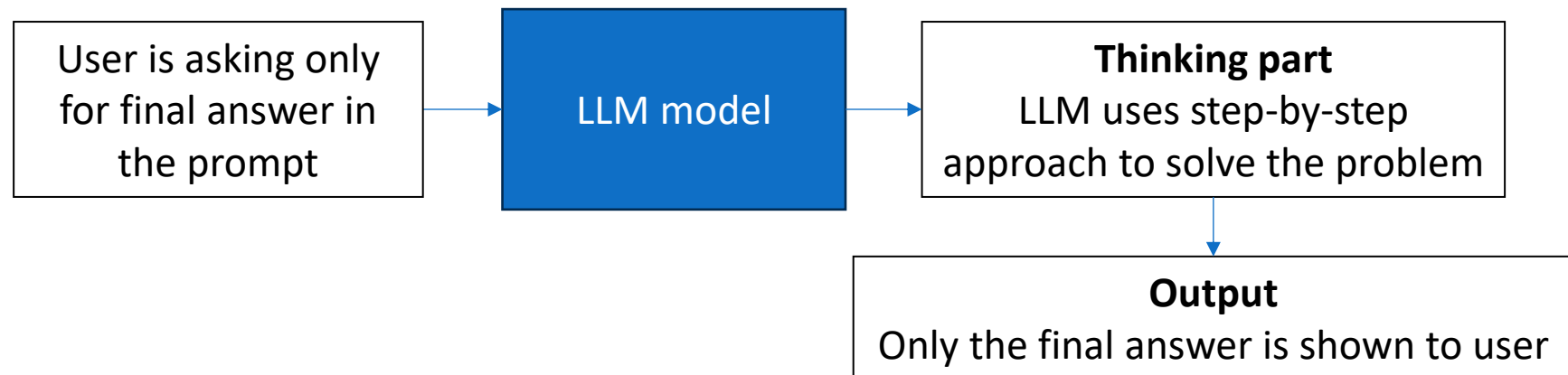
Chain-of-Thought prompting

Reasoning models

Reasoning models always perform CoT automatically

Reasoning models can do two things:

- Apply prompting techniques themselves even if not mentioned in prompt
- They can “think”



Chain-of-Thought prompting

Thinking template

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think>` `</think>` and `<answer>` `</answer>` tags, respectively, i.e., `<think>` reasoning process here `</think>` `<answer>` answer here `</answer>`. User: **prompt**. Assistant:

Table 1 | Template for DeepSeek-R1-Zero. **prompt** will be replaced with the specific reasoning question during training.

Research Paper: <https://arxiv.org/pdf/2501.12948>



Chain-of-Thought prompting

Do we need CoT

When using reasoning models, CoT prompting is not needed

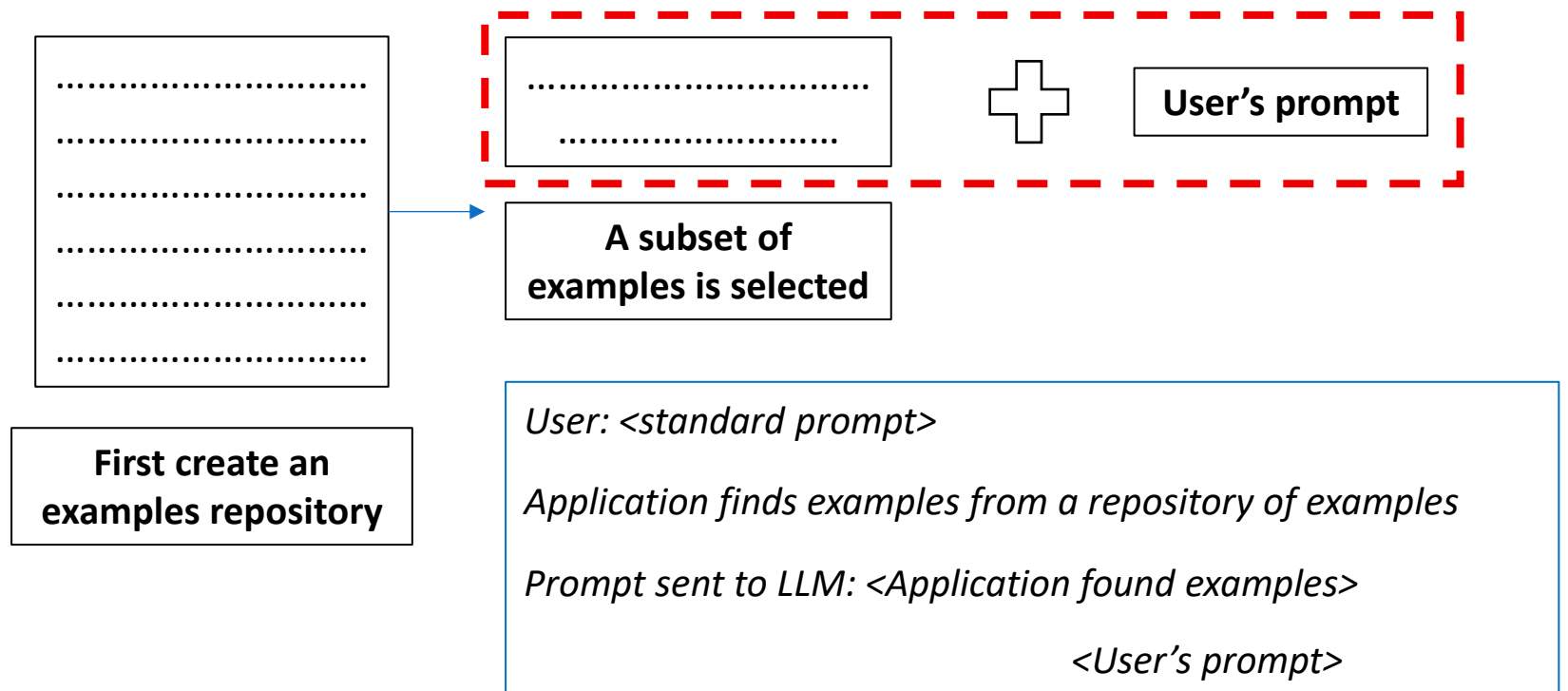
For other models or smaller locally running models, CoT should be used



Chain-of-Thought prompting

Automatic CoT

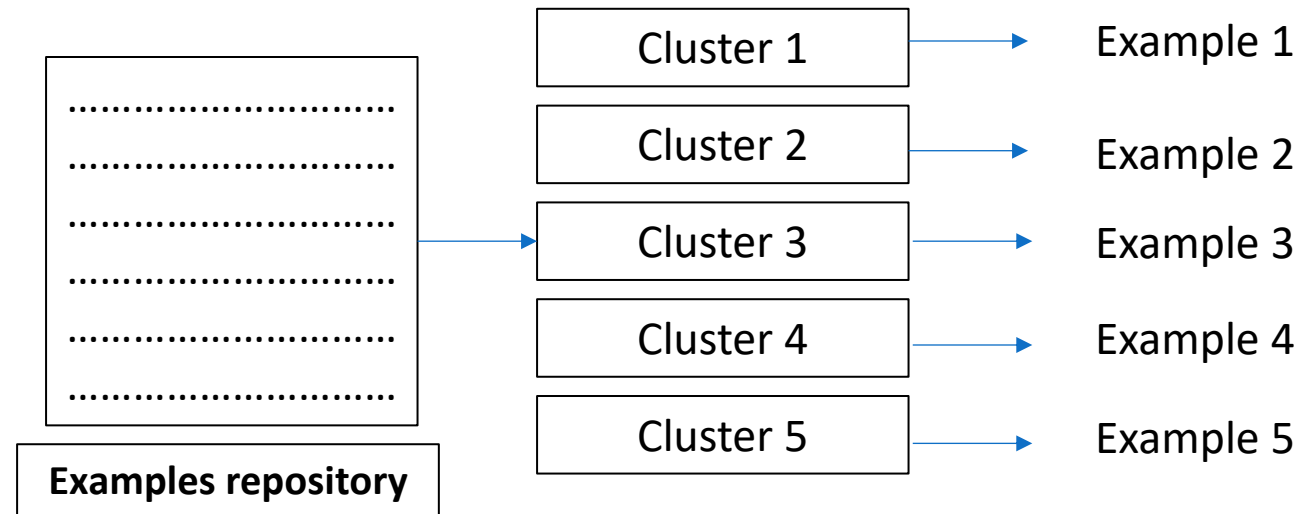
Automatic Chain-of-thought is like few-shots CoT, only difference is that “shots” or examples are not manually given. They are picked from a collection of examples.



Chain-of-Thought prompting

Automatic CoT

Diverse examples in the prompt lead to better reasoning performance



Chain-of-Thought prompting

Automatic CoT

Model	Arithmetic						Commonsense		Symbolic	
	MultiArith	GSM8K	AddSub	AQuA	SingleEq	SVAMP	CSQA	Strategy	Letter	Coin
Zero-Shot	22.7	12.5	77.0	22.4	78.7	58.8	72.6	54.3	0.2	53.8
Zero-Shot-CoT	78.7	40.7	74.7	33.5	78.7	63.7	64.6	54.8	57.6	91.4
Few-Shot	33.8	15.6	83.3	24.8	82.7	65.7	79.5	65.9	0.2	57.2
Manual-CoT	91.7	46.9	81.3	35.8	86.6	68.9	73.5	65.4	59.0	97.2
Auto-CoT	92.0	47.9	84.8	36.5	87.0	69.5	74.4	65.4	59.7	99.9

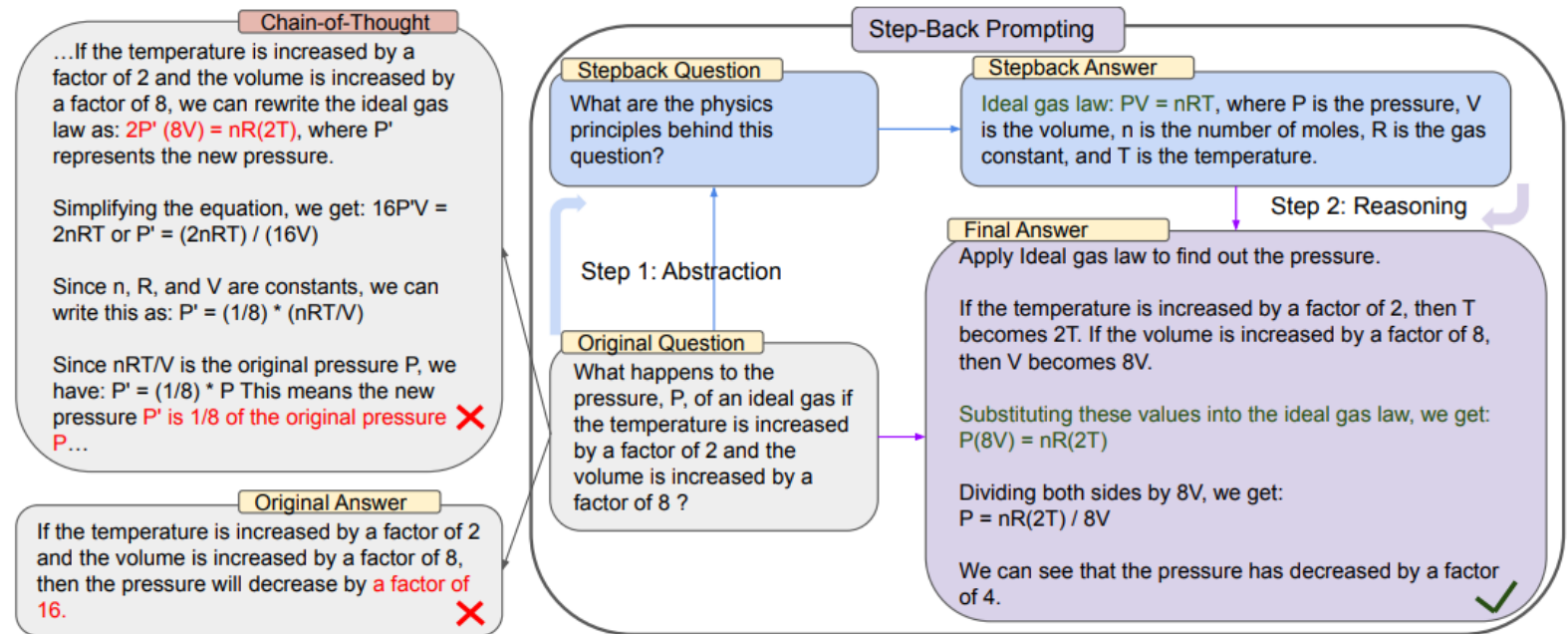


Step-back prompting

What

Similar to CoT, but the first step is a step-back

Step-back – Generate more abstract and general information about the topic



Step-back prompting

Template

Answer this question: <Question>

Let's think step by step to answer this question:

Step 1: Abstract the higher-level concept and principles relevant to this question:

Step 2: Use the abstractions to reason through the question:

Final Answer:



Role prompting

What

Assign the role of expert of the field to the LLM

SEO prompts

- I'm writing an article about 'Run local LLM models using Ollama' What would be the best SEO-friendly URL structure?
- Rewrite the article in an NLP-friendly way: [content section]

Added "You are an SEO expert"

It had NO impact on the quality of output



Role prompting

When it works?

Standard prompt

I see error 2P11 on my canon printer. What does it mean and how to resolve it?

Role prompt

You a customer service executive working for canon. I see error 2P11 on my canon printer. What does it mean and how to resolve it?

Solution presentation was very different

You are a physicist, explain gravity to me

vs

You are a primary grade teacher, explain gravity to me.

Prompt: I'm writing an article about 'Run local LLM models using Ollama' What would be the best SEO-friendly URL structure?



Role prompting

When it works?

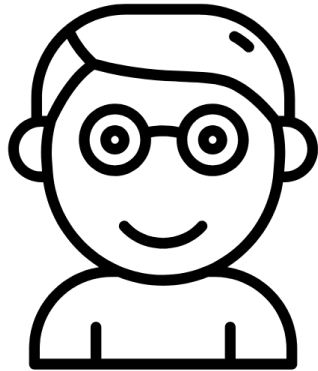
Works best when:

Assigned roles have character and approach the problem differently (positively) than others



Self-consistency

When it works?



$$3x - 5y = 1$$

$$x + 2y = 4$$

Find value of x and y

Solution 1

$$3x - 5y = 1$$

$$3x + 6y = 12$$

$$11y = 11$$

$$y = 1$$

$$x = 2$$



Solution 2

$$x = 4 - 2y$$

$$3(4 - 2y) - 5y = 1$$

$$12 - 11y = 1$$

$$y = 1$$

$$x = 2$$



Self-consistency

Prompt Template

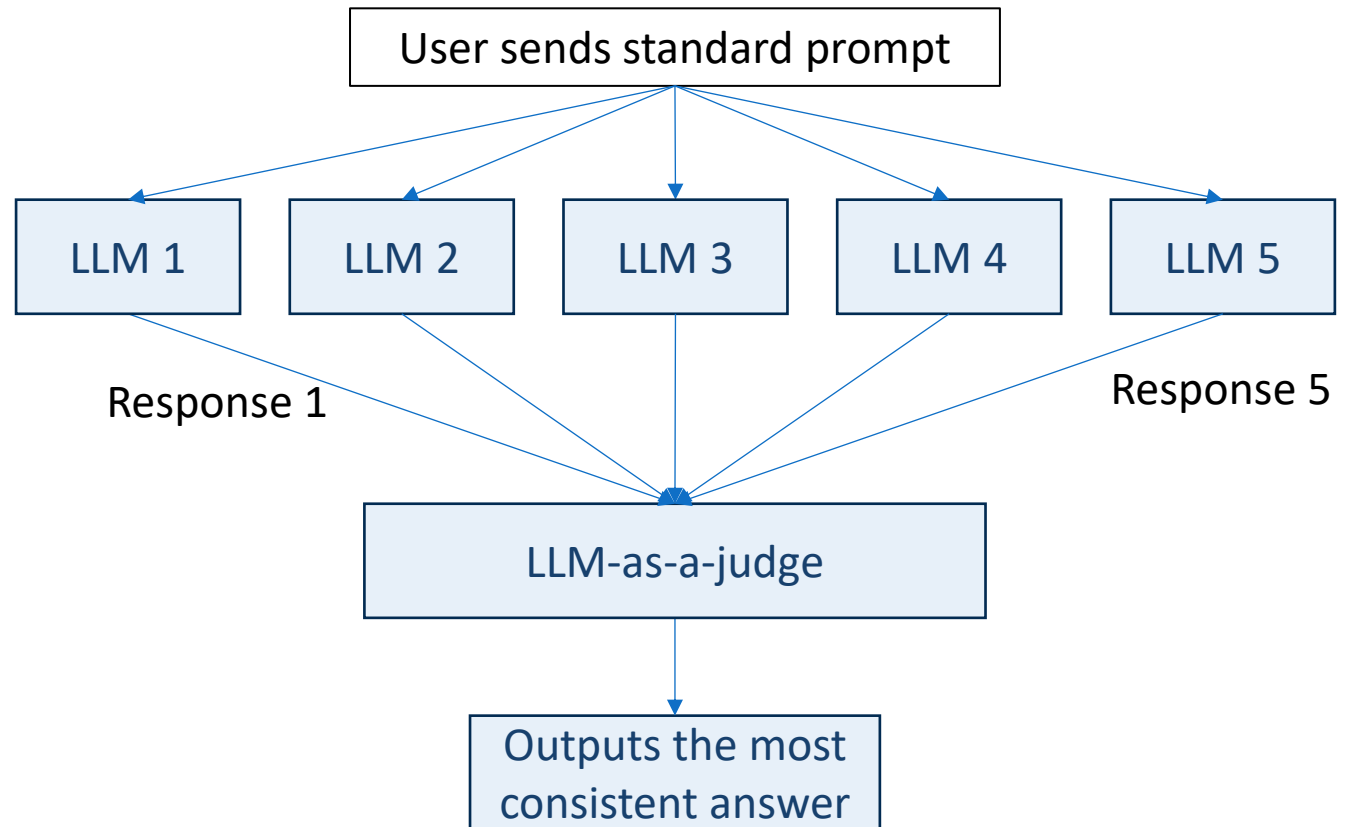
For solving the following question, first generate multiple reasoning paths and aggregate the final answers to come to a final conclusion.

Question: <Question>

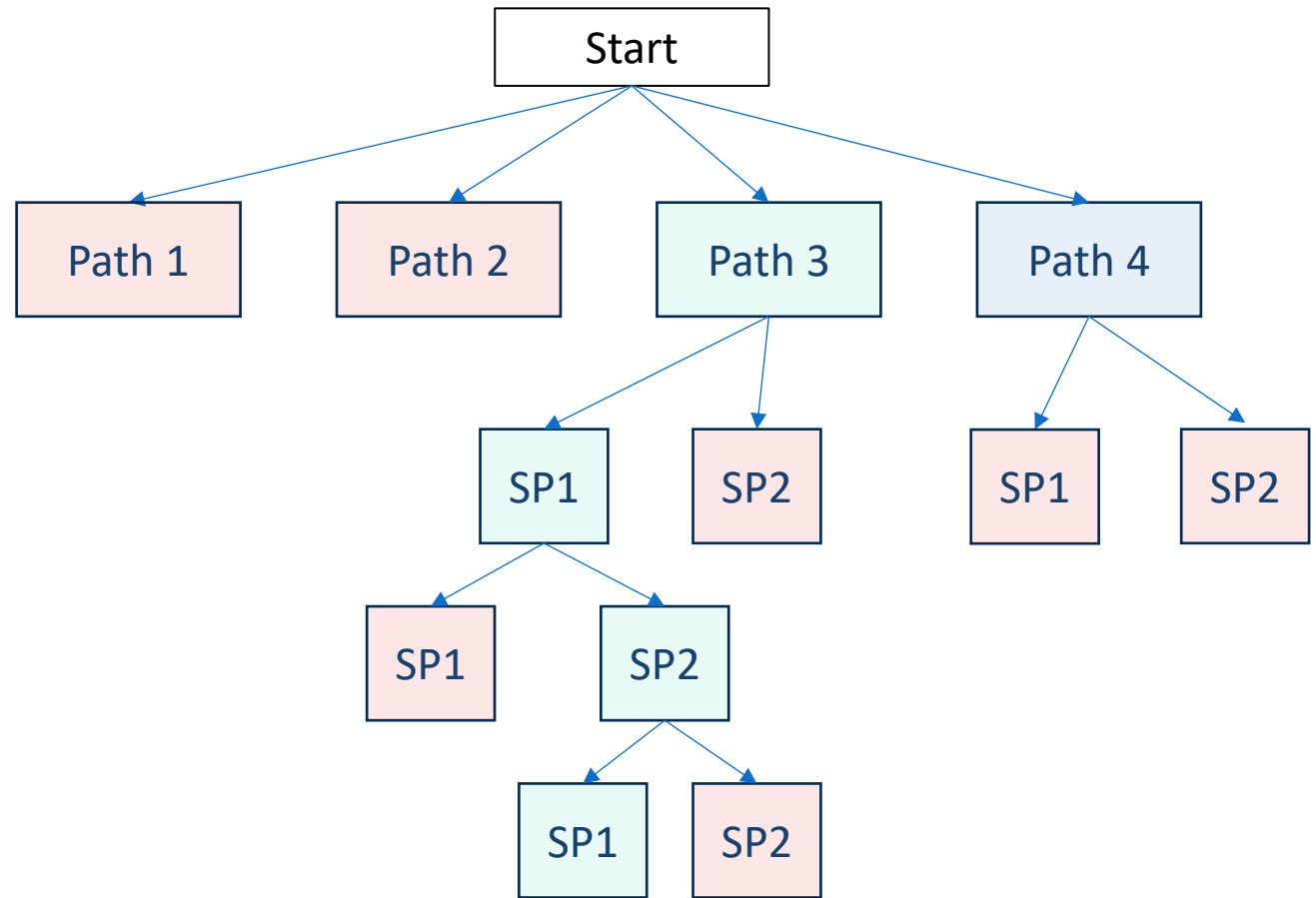
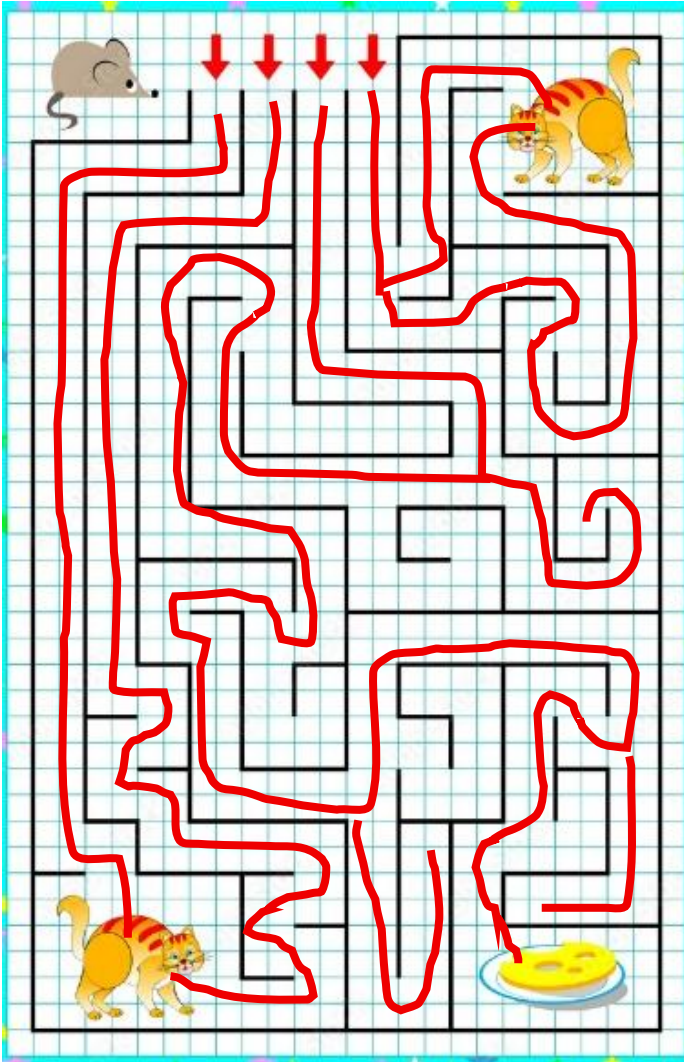


Self-consistency

Using code

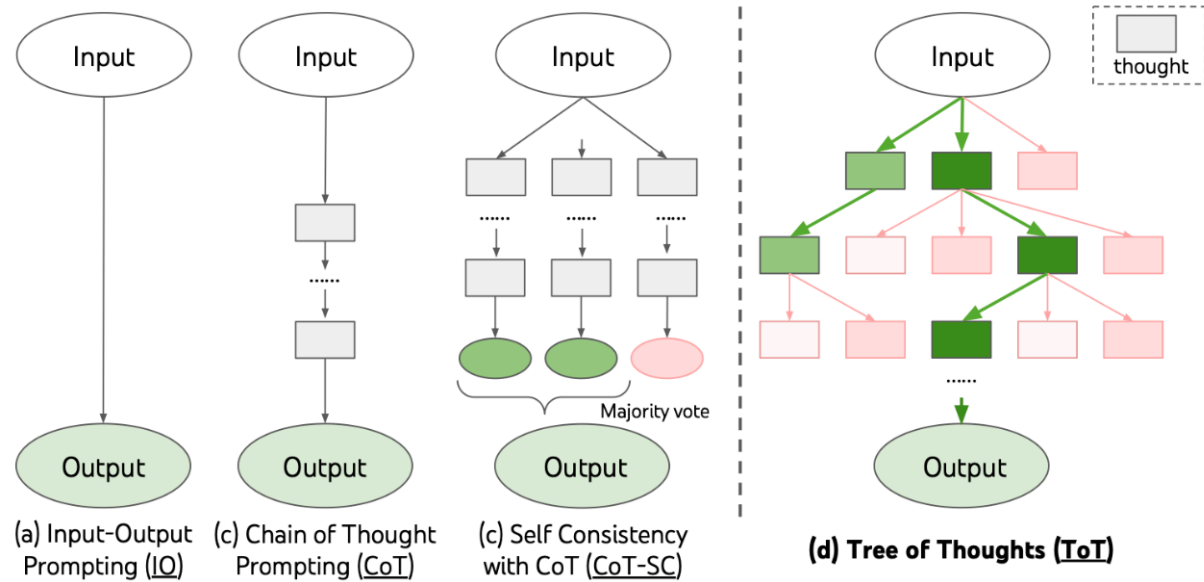


Tree-of-thought



Tree-of-thought

Comparison



Tree-of-thought research paper: <https://arxiv.org/pdf/2305.10601>



Tree-of-thought

Comparison

Method	Success
IO prompt	7.3%
CoT prompt	4.0%
CoT-SC (k=100)	9.0%
ToT (ours) (b=1)	45%
ToT (ours) (b=5)	74%
IO + Refine (k=10)	27%
IO (best of 100)	33%
CoT (best of 100)	49%

Table 2: Game of 24 Results.

Tree-of-thought research paper: <https://arxiv.org/pdf/2305.10601>



Tree-of-thought

Prompt

Imagine three different experts are answering this question.

All experts will write down 1 step of their thinking, then share it with the group.

Then all experts will go on to the next step, etc.

If any expert realises they're wrong at any point, then they leave.

The question is...

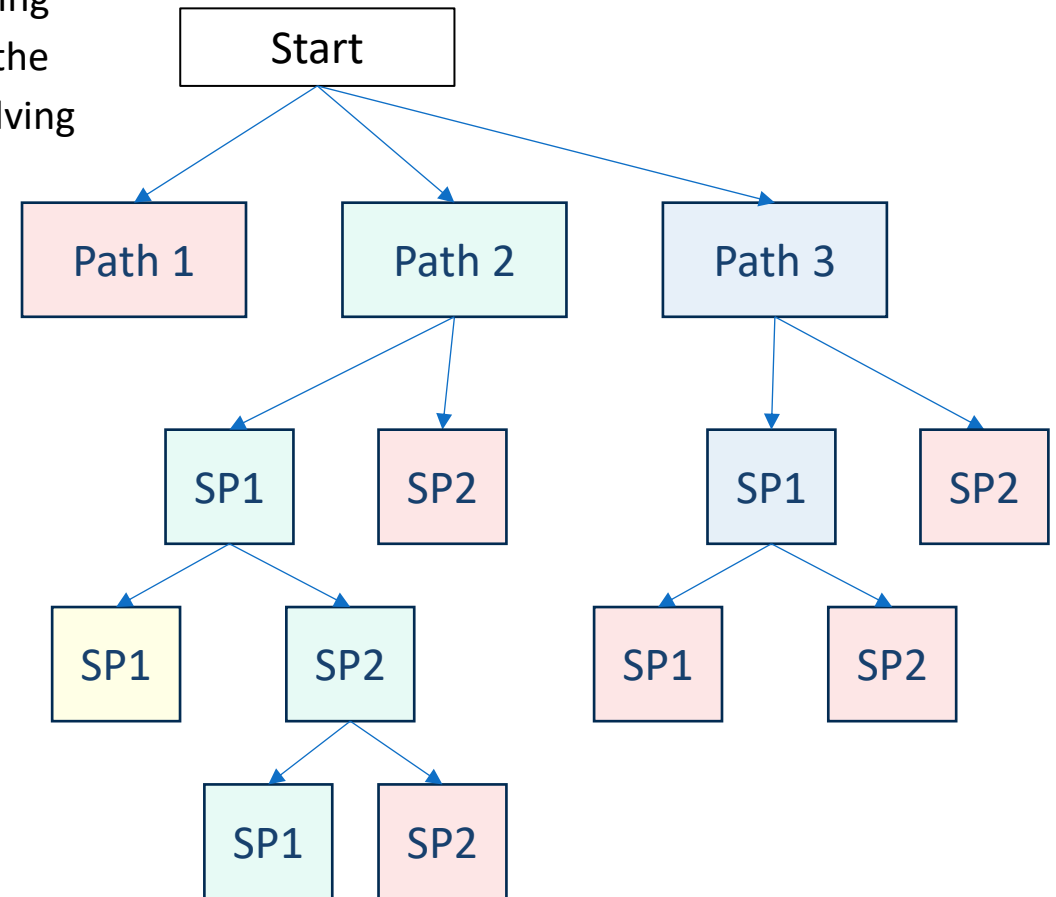
Source: <https://github.com/dave1010/tree-of-thought-prompting>



Tree-of-thought

Evaluate multiple ways of solving this problem and write down the first step of all these ways of solving

Depth First Search (DFS)



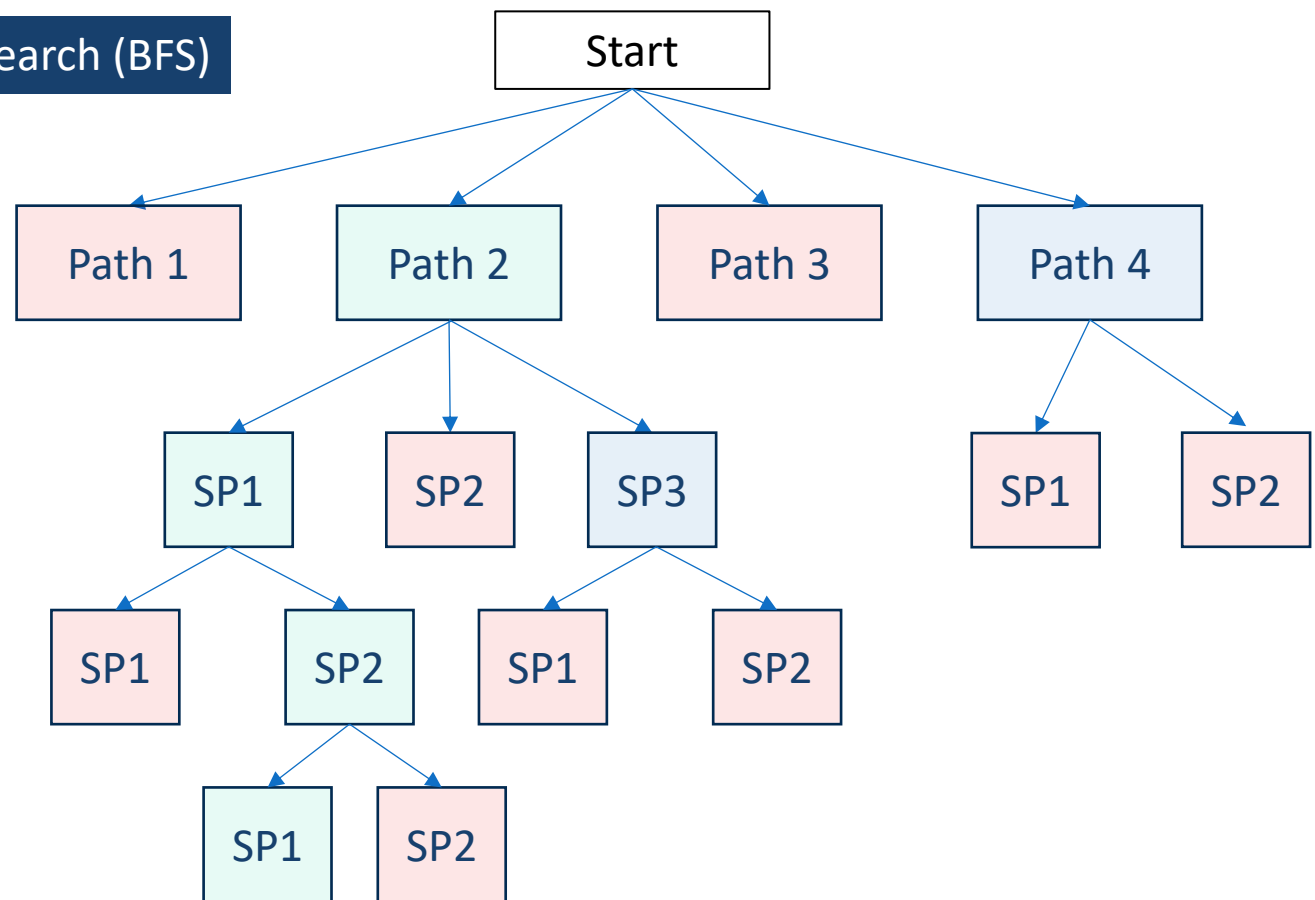
Types of
implementations



Tree-of-thought

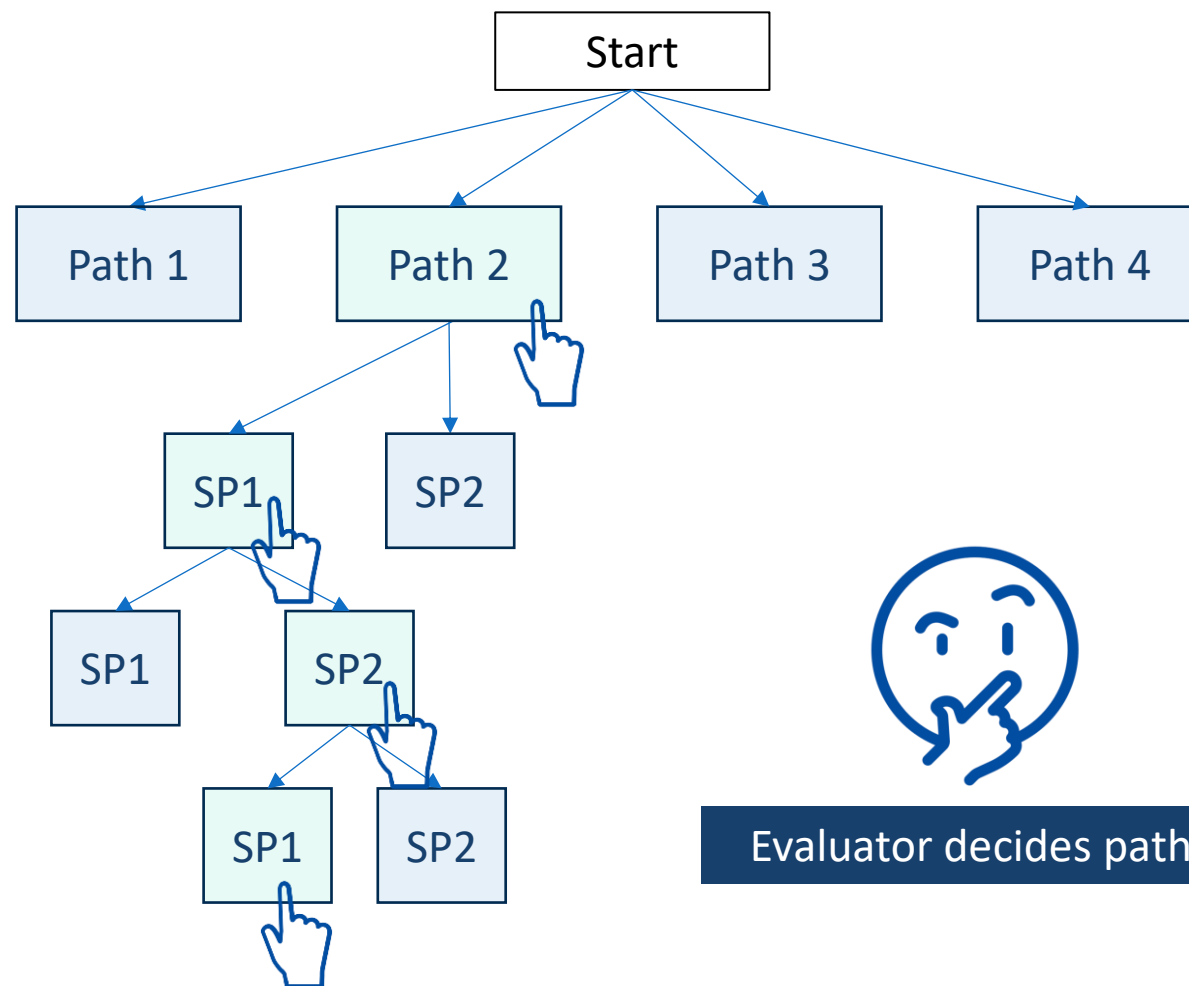
Types of implementations

Breadth First Search (BFS)



Tree-of-thought

Types of implementations



Tree-of-thought

Drawback

Tree-of-thought is token intensive – response can be 100x expensive



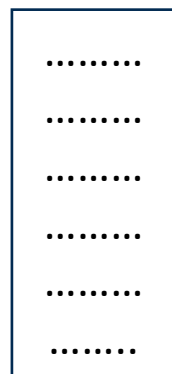
Skeleton-of-thought

Scenario

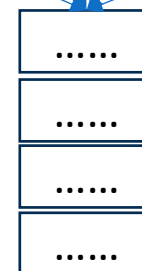
Write a 10-page essay



Option 1



Option 2



Skeleton-of-thought

Steps

Step 1: Generating the skeleton

Prompt: Create a skeleton of the key-points to answer the question
<Question>

Step 2: Parallel expansion on the generated points

Prompt: Question: <Question>

The answer has the following skeleton: <Skeleton>

Your task is to expand the point <point number and text> only. Write 5 sentences. Do not expand other points.

Send this in parallel requests

Step 3: Collating the responses

Output: Here's an essay on "topic"

<response 1> <response 2> <response 3> <response 4>...

Reference Paper: <https://arxiv.org/pdf/2307.15337>



Skeleton-of-thought

Benefit

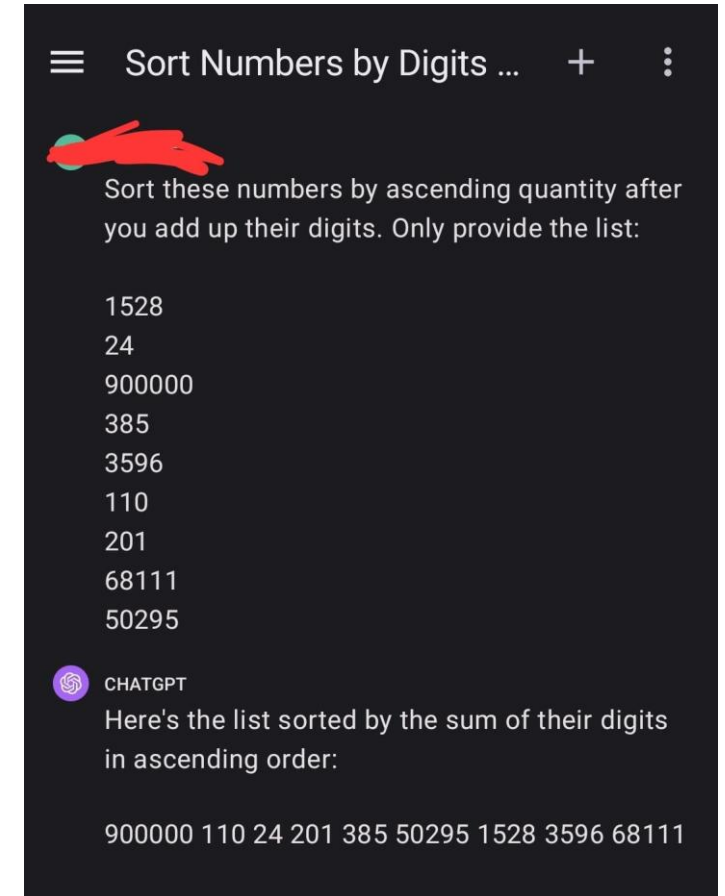
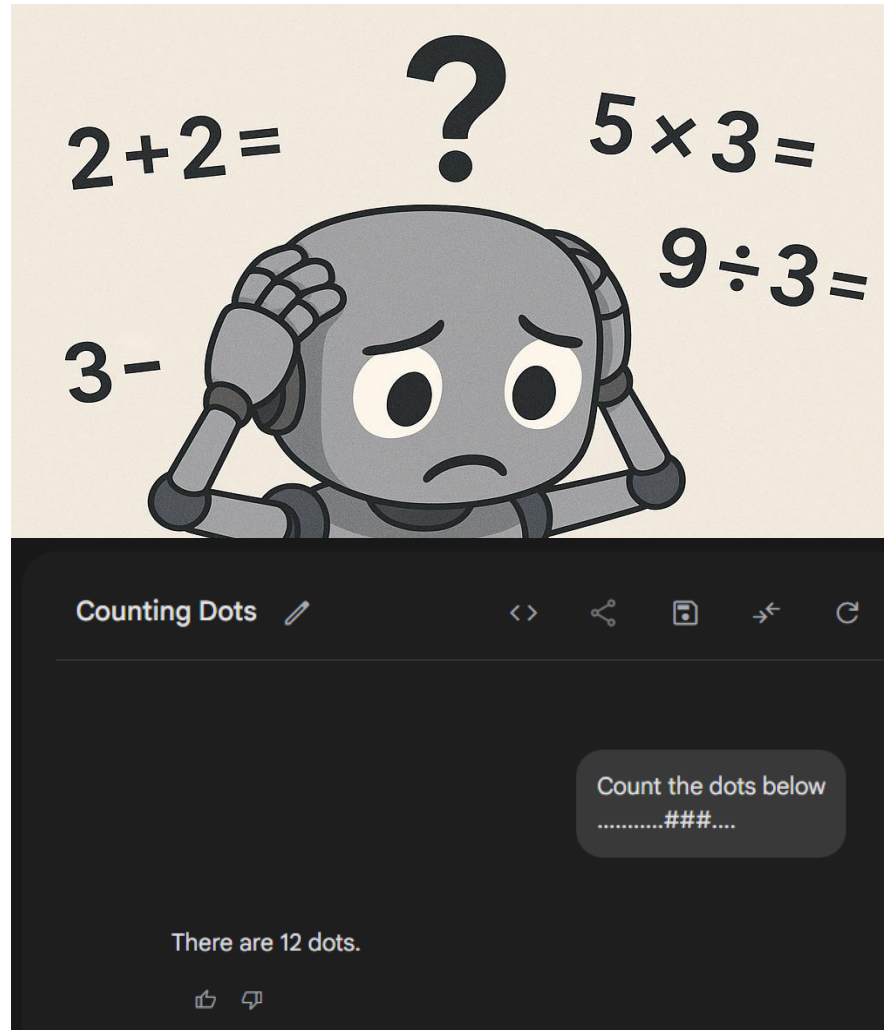
Low latency

Best for generating huge textual response in less time



Program-of-thought

Math problems



Program-of-thought

Solution

LLM's are good at writing code



Program-of-thoughts

Write code corresponding to the problem



Run the code to get the result

Reference Paper: <https://arxiv.org/pdf/2211.12588>



Program-of-thought

LLM capabilities

- Identify the given task is a math problem
- Write and run code to answer such question



Program-of-thought

Template

Question: <Question>

Answer this question by implementing a solver() function.

def solver():

Lets write a python code step by step, and then return the answer

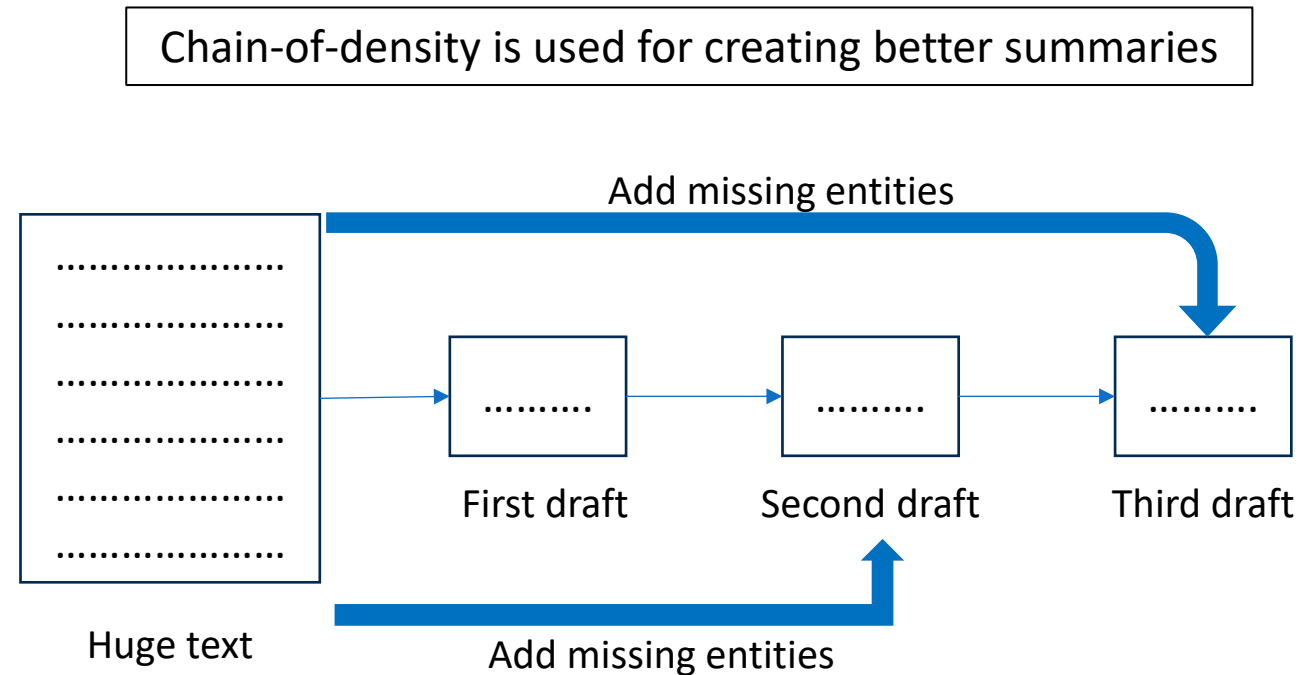
Firstly, lets define the variables:

Few shot technique can also be used



Chain-of-Density

Concept



Entity is defined as :

Relevant: to the main story.

Specific: descriptive yet concise

Novel: not in the previous summary

Faithful: present in the Article

Anywhere: located anywhere in the Article.



Chain-of-Density

Template

Article:

###

<Article>

###

Summary:

<Summary>

Task:

###

Identify 2 informative entities from the article which are missing from this summary. An entity is defined as:

Relevant: to the main story.

Specific: descriptive yet concise

Novel: not in the previous summary

Faithful: present in the Article

Anywhere: located anywhere in the Article.

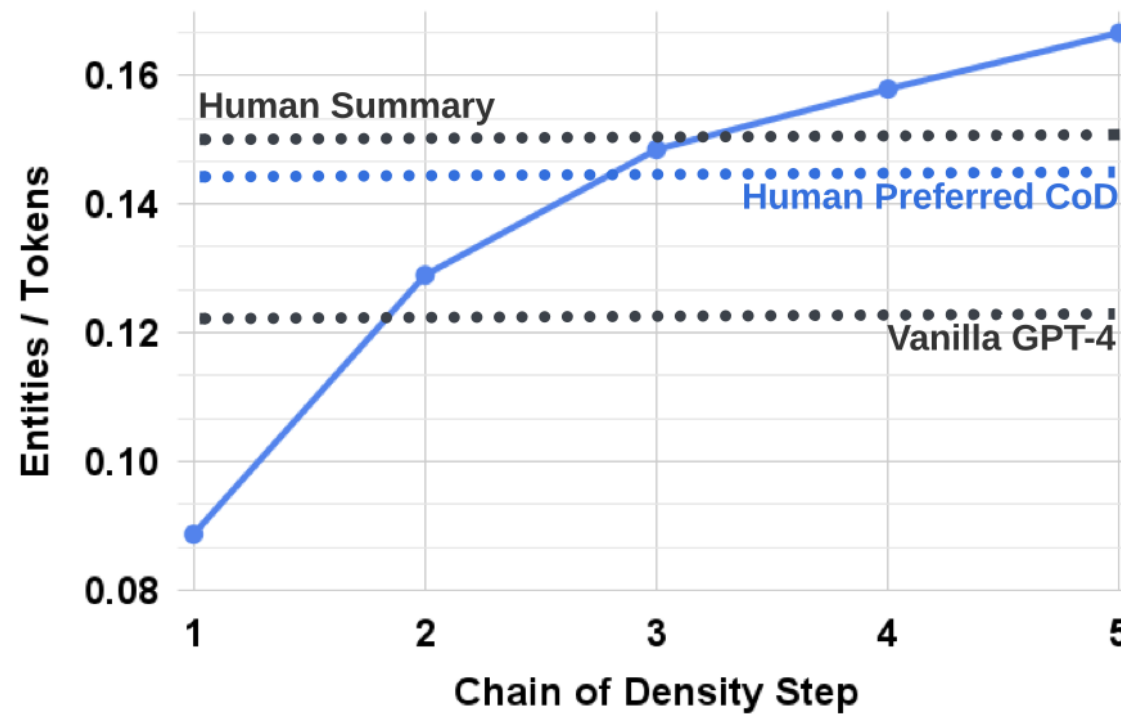
Then, write a new denser summary of identical length which covers all entities and details from previous summary plus the missing entities.

###



Chain-of-Density

Results



Hyperparameters

What

Prompt hyperparameters, not LLM training hyperparameters

Prompt hyperparameters are additional parameters that can be sent with the prompt and affect the response from LLM

Hyperparameter
Temperature
Top_p
Max_tokens
Presence_penalty
Frequency_penalty
Stop sequences

1. Understanding these parameters
2. Optimum value range
3. Hyperparameter tuning



Hyperparameters

How LLMs work?

Sky

is

Blue

Clear

Cloudy

Green

Logits

3

2

1

0.5

Probability

63%

23%

9%

5%



Hyperparameters work here



Hyperparameters

Logits

	Logits		Probability
Blue	3	$P_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$	63%
Clear	2		23%
Cloudy	1		9%
Green	0.5		5%



Hyperparameters

Temperature

- Varies between 0 and 2
- Lower value - deterministic and less varied output
- Higher value - more diverse and creative output

$$\text{Adjusted Logit}_i = \frac{\text{Raw Logit}_i}{T}$$



Hyperparameters

Temperature

Logits

Blue

3

Clear

2

Cloudy

1

Green

0.5

Next Token	Raw Logit	$T = 0.5 (\text{logit}/0.5)$	$T = 1.0 (\text{logit}/1)$	$T = 2.0 (\text{logit}/2)$
blue	3	6	3	1.5
clear	2	4	2	1
cloudy	1	2	1	0.5
green	0.5	1	0.5	0.25



Hyperparameters

Temperature

T = 1.0 (Standard probabilities)

Next Token	Adjusted Logit	e^(logit)	Probability
blue	3	20.09	63%
clear	2	7.39	23%
cloudy	1	2.72	9%
green	0.5	1.65	5%

T = 0.5 (Lower temp → More confident prediction)

Next Token	Adjusted Logit	e^(logit)	Probability
blue	6	403.43	86%
clear	4	54.6	12%
cloudy	2	7.39	2%
green	1	2.72	1%

T = 2.0 (Higher temp → Less confident, more diverse)

Next Token	Adjusted Logit	e^(logit)	Probability
blue	1.5	4.48	44%
clear	1	2.72	27%
cloudy	0.5	1.65	16%
green	0.25	1.28	13%

$$P_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$



Hyperparameters

Temperature

- Higher value - more diverse and creative output – Writing jobs like essays, poems etc
- Lower value - deterministic and less varied output – Factual Q/A

Hyperparameter	Creative & Diverse Tasks	Deterministic & Consistent Tasks
Temperature	0.7 – 1.2	0.0 – 0.7

If $T=0$, model picks only the word with top probability



Hyperparameters

Top-p

Next Token	Adjusted Logit	$e^{(\text{logit})}$	Probability	Cumulative Probability
blue	3	20.09	63%	63%
clear	2	7.39	23%	86%
cloudy	1	2.72	9%	95%
green	0.5	1.65	5%	100%

If Top-p = 0.95, only consider tokens with cumulative probability $\geq 95\%$

If Top-p = 0.5, only “blue” will be considered for next token

Higher value – more options to select from - more diverse and creative

Lower value – less options to select from - deterministic and less varied



Hyperparameters

Top-p

$T = 1.0$, $\text{Top-p} = 0.95$

Next Token	Adjusted Logit	$e^{\text{(logit)}}$	Probability	Cumulative Probability
blue	3	20.09	63%	63%
clear	2	7.39	23%	86%
cloudy	1	2.72	9%	95%
green	0.5	1.65	5%	100%

$T = 0.5$, $\text{Top-p} = 0.95$

Next Token	Adjusted Logit	$e^{\text{(logit)}}$	Probability	Cumulative Probability
blue	6	403.43	85%	85%
clear	4	54.6	12%	97%
cloudy	2	7.39	2%	99%
green	1	2.72	1%	100%



Hyperparameters

Top-p

Suggested values:

Hyperparameter	Creative & Diverse Tasks	Deterministic & Consistent Tasks
Top_p (nucleus)	0.9 – 1.0	0.5 – 0.9



Hyperparameters

Max tokens & Stop sequence

Max tokens is a check on the numbers of tokens generated

Does not instruct LLM on size of response, it is a hard stop if response is going beyond

Stop sequence is a set of tokens that if LLM is about to print, it stops immediately



Hyperparameters

Presence & Frequency penalty

Encourages variety

Penalty on usage of same words again in the response

Adjusted logit = Original Logit - (presence_penalty if token appeared)

Adjusted logit = Original Logit - (frequency_penalty × occurrences)

If “relativity” as already come twice in the text,

With presence penalty = 1, Adjusted logit = Original logit -1

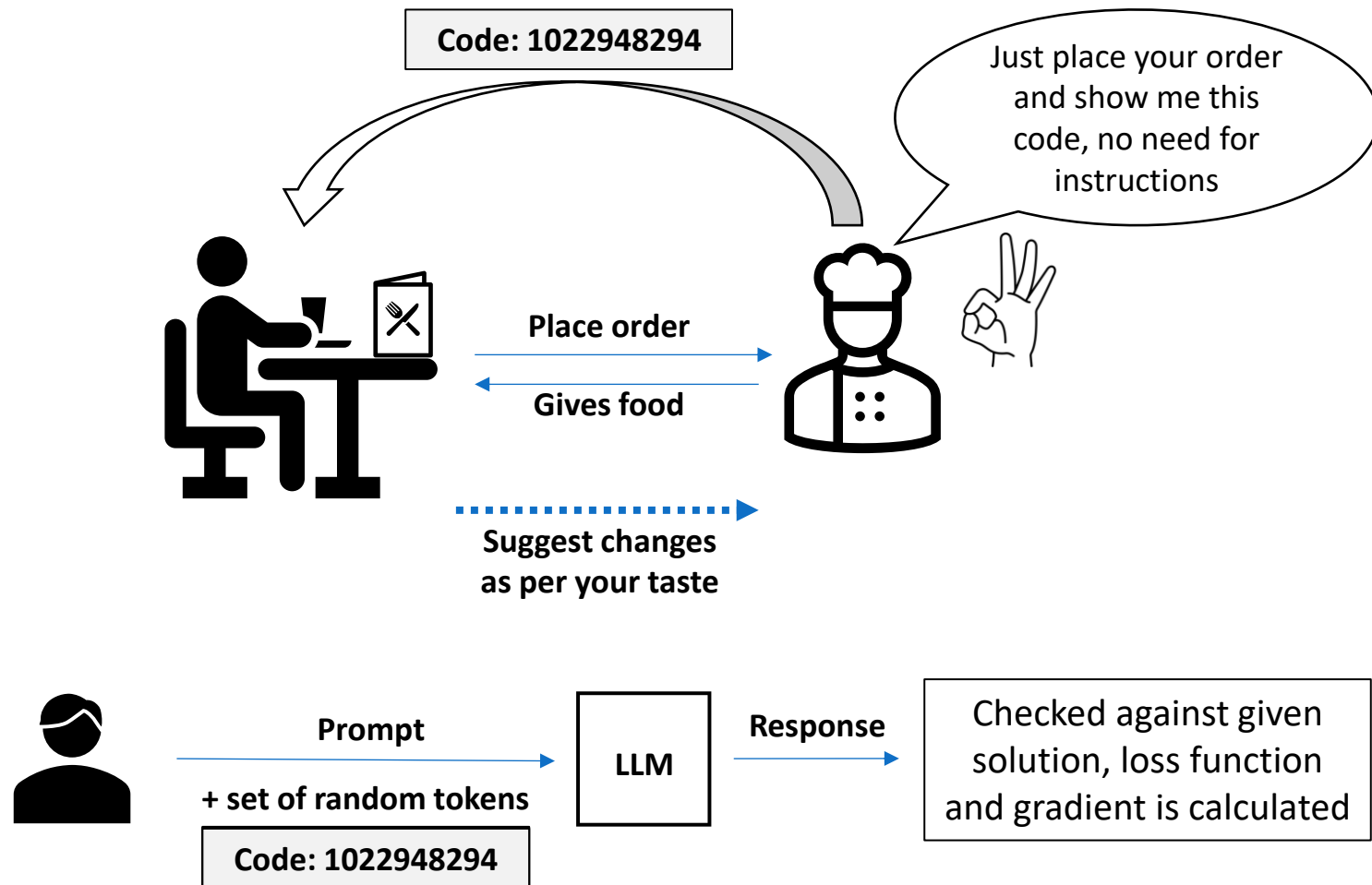
With frequency penalty = 1, Adjusted logit = Original logit - 2

Presence penalty has more impact for more occurrence, frequency penalty has same impact regardless of number of occurrences



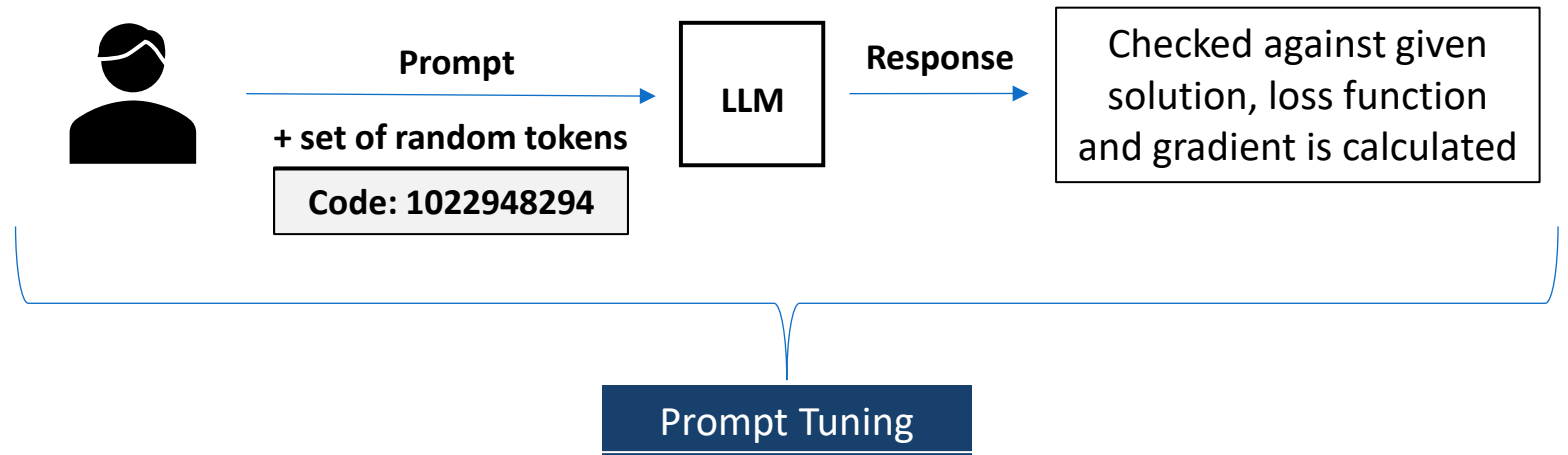
Prompt Fine-tuning

Scenario



Prompt Fine-tuning

Terminologies



This is not LLM fine-tuning. Prompt tuning is simpler less resource heavy

Set of tokens found after tuning process is called the **soft prompt**

User's inputted prompt is called the **hard prompts**



Prompt Fine-tuning

Prompt vs Model tuning

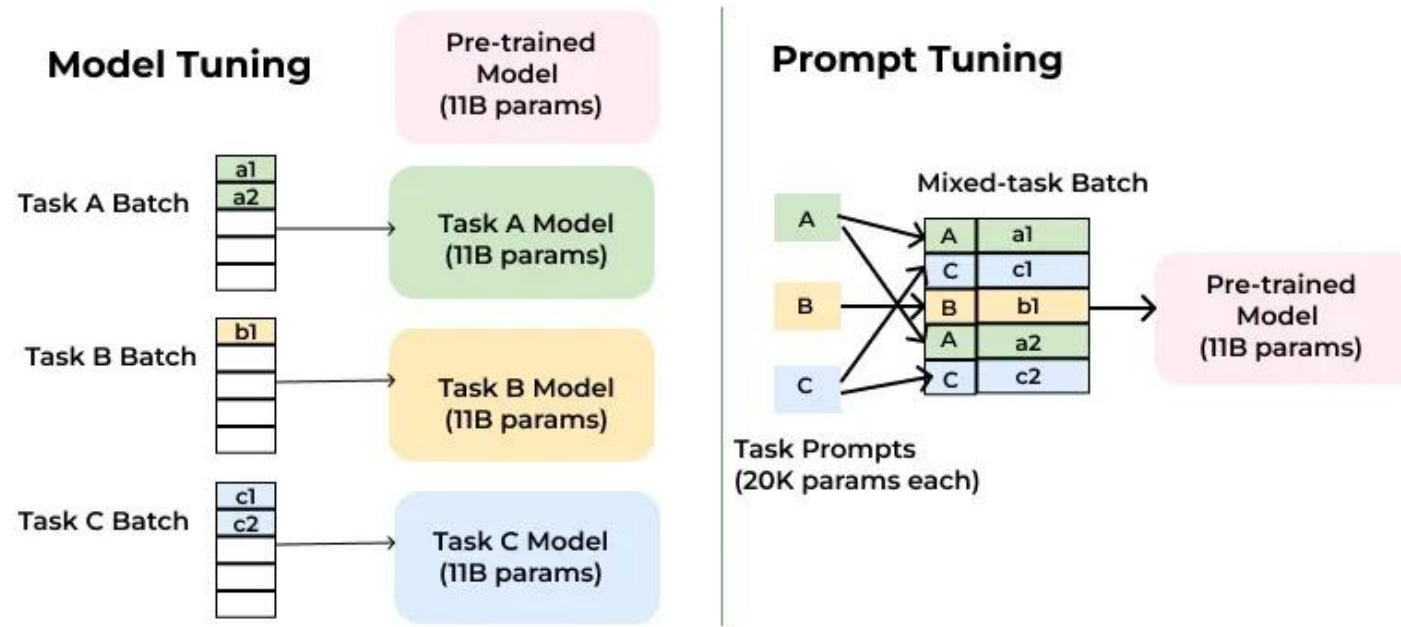


Image from this paper: <https://aclanthology.org/2021.emnlp-main.243.pdf>



Prompt Fine-tuning

Process

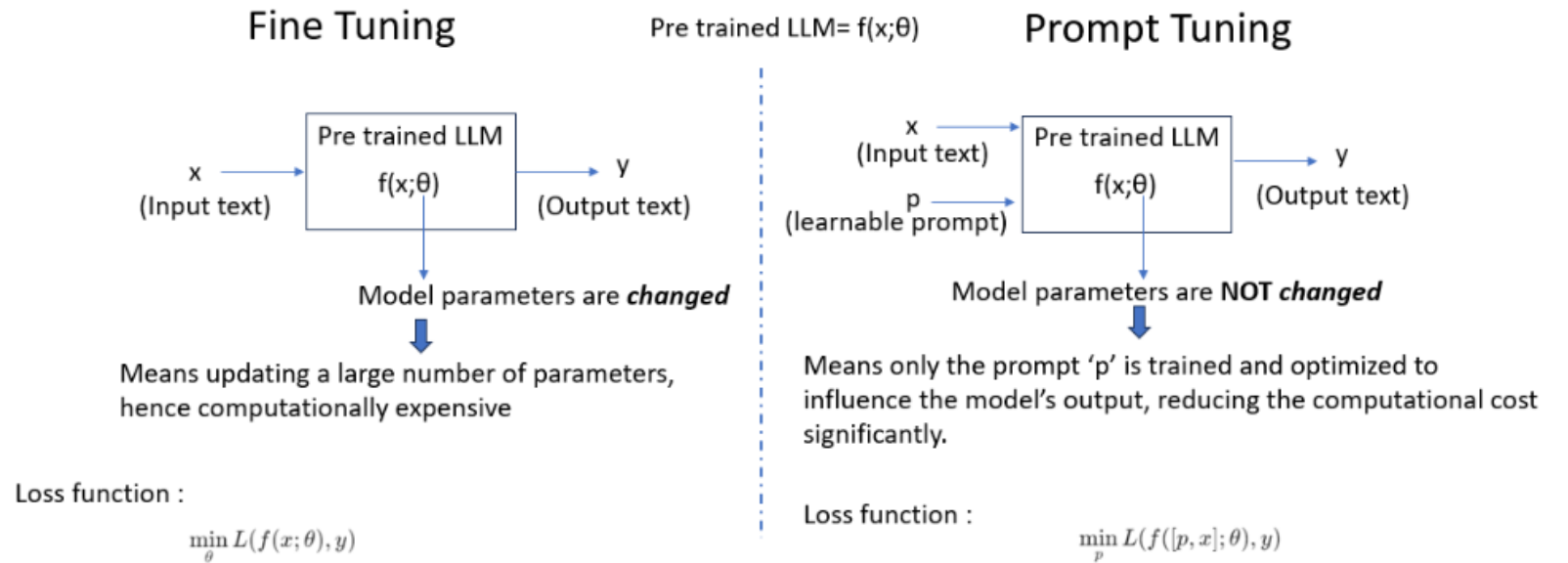


Image from this source: <https://developer.ibm.com/articles/awb-how-prompt-tuning-works/>



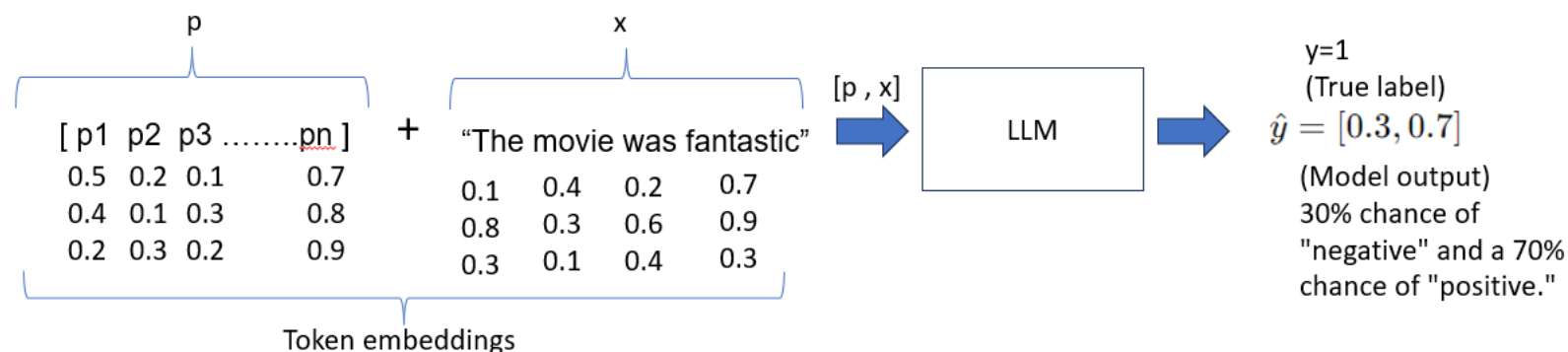
Prompt Fine-tuning

Process

Task: We want to perform sentiment analysis. User's review of a movie will be sent to LLM and LLM must categorize it as "positive" or "negative"

What we have: Dataset of customer reviews with the label "positive" or "negative"

Instead of sending the prompt "Classify the sentiment of a given text as positive or negative", we are going to send a set of trainable parameters **p**



Example from this source: <https://developer.ibm.com/articles/awb-how-prompt-tuning-works/>



Prompt Fine-tuning

Process

Loss function: based on difference between predicted and actual output

$$L = -y \log(\hat{y}_{\text{positive}}) - (1 - y) \log(\hat{y}_{\text{negative}})$$

With $y=1$ and $\hat{y}_{\text{positive}} = 0.7$

The the loss function result is $L=-(1)\log(0.7)=0.357$

Gradient descent: Update the prompt embeddings $\mathbf{p}$ based on the gradients of the loss function with respect to $\mathbf{p}$

$$\mathbf{p} \leftarrow \mathbf{p} - \eta \nabla_{\mathbf{p}} L$$

Where:

η is the learning rate (a small step size that controls how much we adjust the prompt).

$\nabla_{\mathbf{p}} L$ is the gradient of the loss with respect to the prompt $\mathbf{p}$.

Example from this source: <https://developer.ibm.com/articles/awb-how-prompt-tuning-works/>



Prompt Fine-tuning

Example code

https://huggingface.co/learn/cookbook/en/prompt_tuning_peft

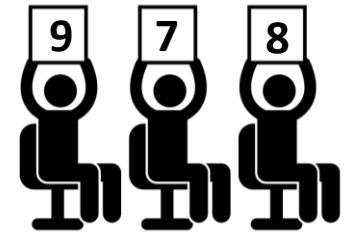
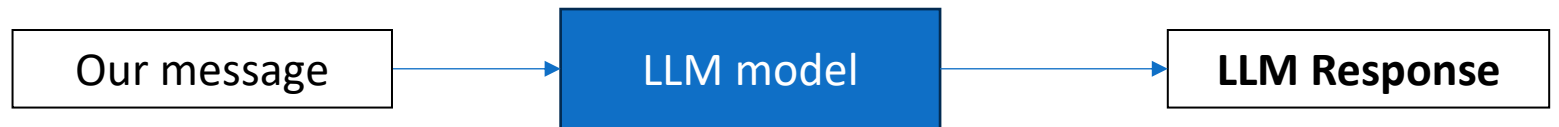
<https://developer.ibm.com/articles/awb-how-prompt-tuning-works/>



Prompt Evaluation

What

Prompt evaluation: Assessing and refining prompts for better outputs



Methods for evaluating prompts:

- Human evaluation
- Automated metrics
- LLM-as-a-judge



Prompt Evaluation

Human Evaluation

People manually review the outputs of a prompt

Human evaluators can capture nuances that automated methods might miss

Gold standard for assessing subjective qualities like creativity, tone, or usefulness

Review

.....
.....
.....

AI Summary

.....



Missing the tone
of the review



Prompt Evaluation

Human Evaluation

Pros

- Very flexible
- Can assess almost any aspect of output quality
- Closely reflects real-world preferences

Cons

- Time-consuming
- Expensive
- Not scalable
- Different people might have different opinions

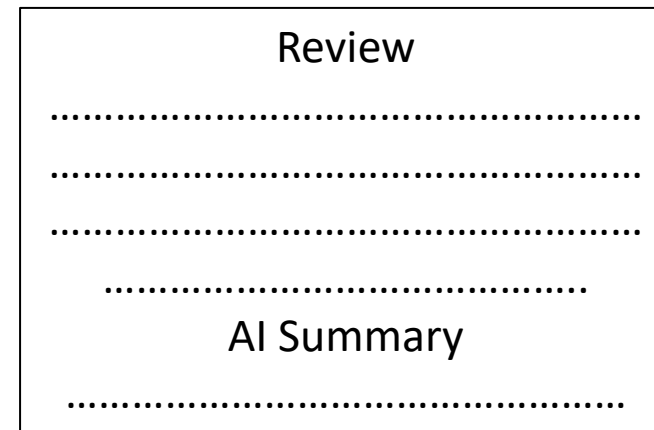
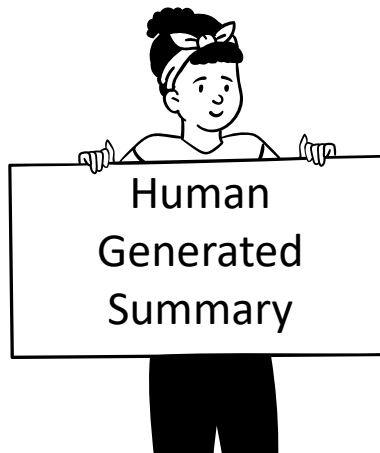


Prompt Evaluation

Automated Metrics

Use algorithms to score the quality of an AI's output

Example: BLEU, ROUGE, or BERTScore for summarization and translation,
Exact match for factual, cosine similarity for embedding space similarity



ROUGE score



Prompt Evaluation

Automated Metrics

Pros

- Fast
- Repeatable
- Scalable

Cons

- Require reference outputs
- Focus on surface-level similarity

Metric like BLEU/ROUGE might not catch nuances like tone, coherence, or logical correctness



Prompt Evaluation

LLM-as-a-judge

Using a Language Model itself to evaluate outputs

LLM acts as a reviewer – it might give a score, choose the better of two outputs, or check if certain criteria are met

Single output scoring (pointwise)

The LLM reviews one output at a time and gives it a score.

Pairwise comparison: The LLM is given two and asked to decide which is better



Prompt Evaluation

LLM-as-a-judge

Pros

- Fast
- Scalable
- Able to handle nuanced criteria

Cons

- Can have biases
- Can be tricked by phrasing
- Additional cost of API calls



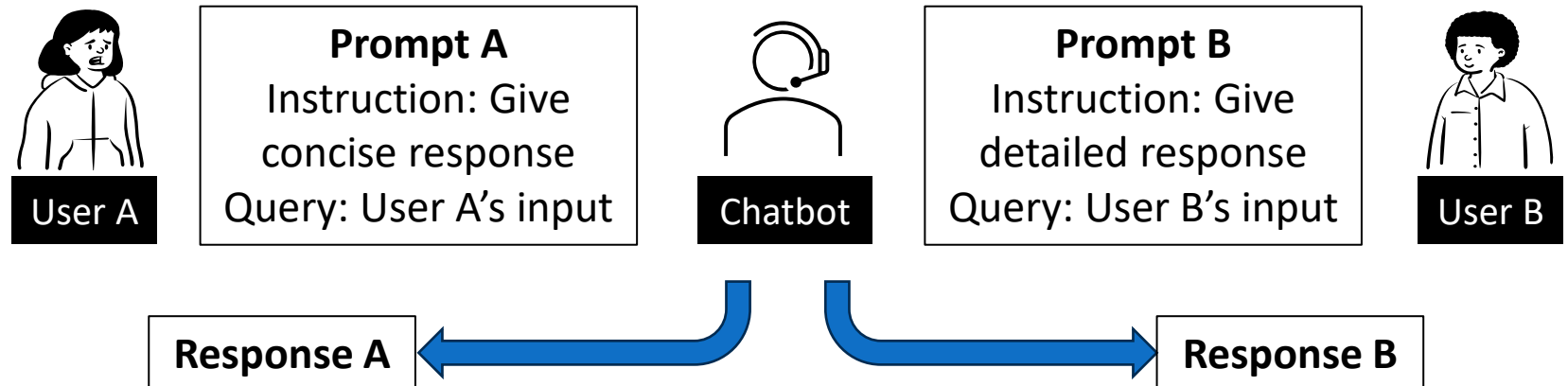
Prompt Evaluation

A/B testing

Comparing two prompts & measuring which performs better on real metrics

Prompt A vs Prompt B

Done on live data – some traffic goes to prompt A and some to prompt B



Collect Metrics

Did the user ask a follow-up question or was their issue resolved?
Did they give a high rating to the chat or leave positive feedback?



Prompt Evaluation

A/B testing

Pros: provides statistical evidence of which prompt is truly better for your application

Challenges

- Requires a deployed system and real
- Risk of exposing a potentially worse prompt to users
- Focuses on overall outcomes; does not tell why one prompt is better

